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MAML-KT: Addressing Cold Start Problem in Knowledge Tracing for New Students via Few-Shot Model-Agnostic Meta Learning

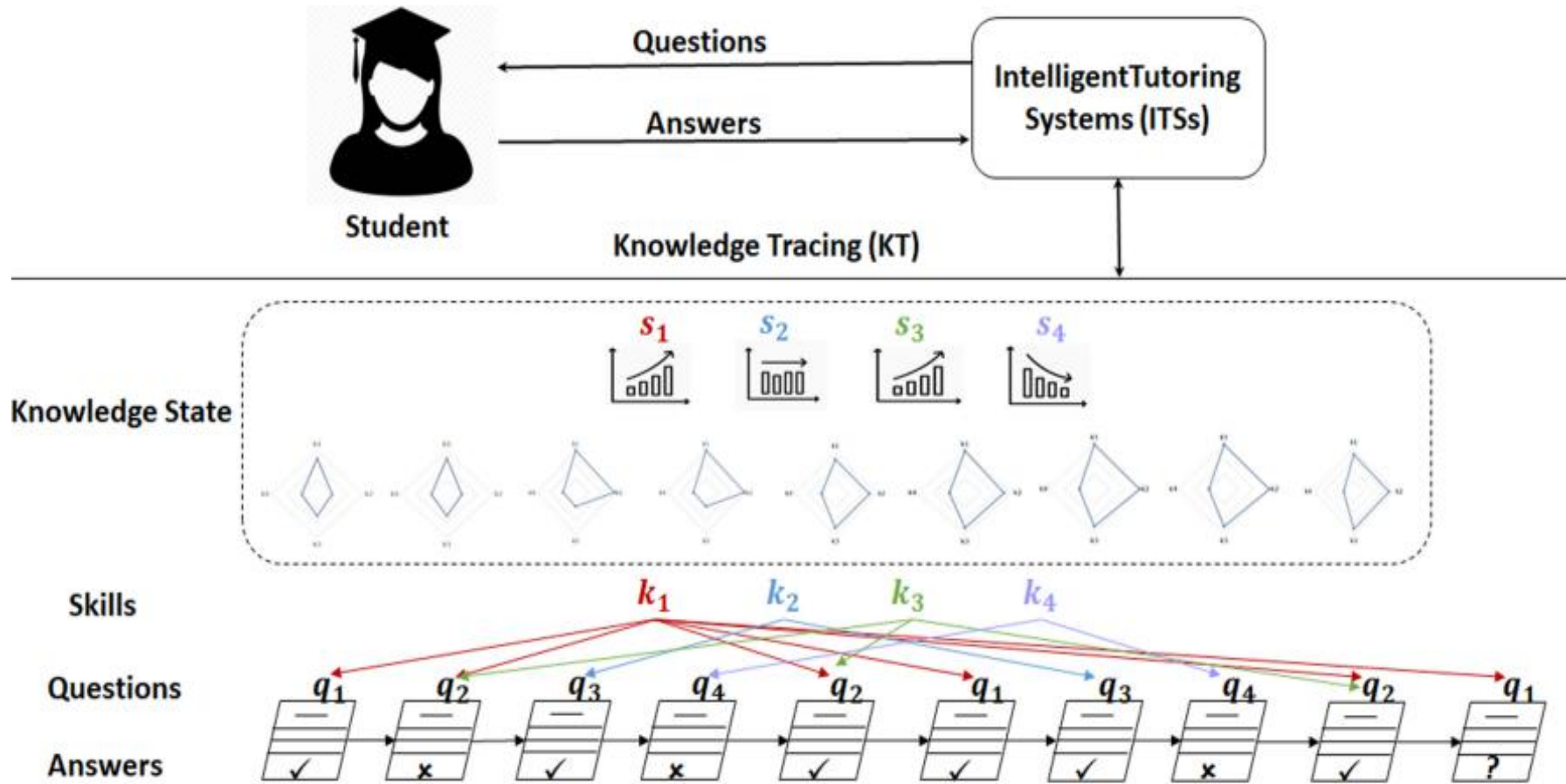
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Overview

- Background
- Intuition for Meta Learning
- MAML-KT
- Research Questions
- Methodology
- Algorithm
- Results
- Future/Ongoing Works
- Conclusion

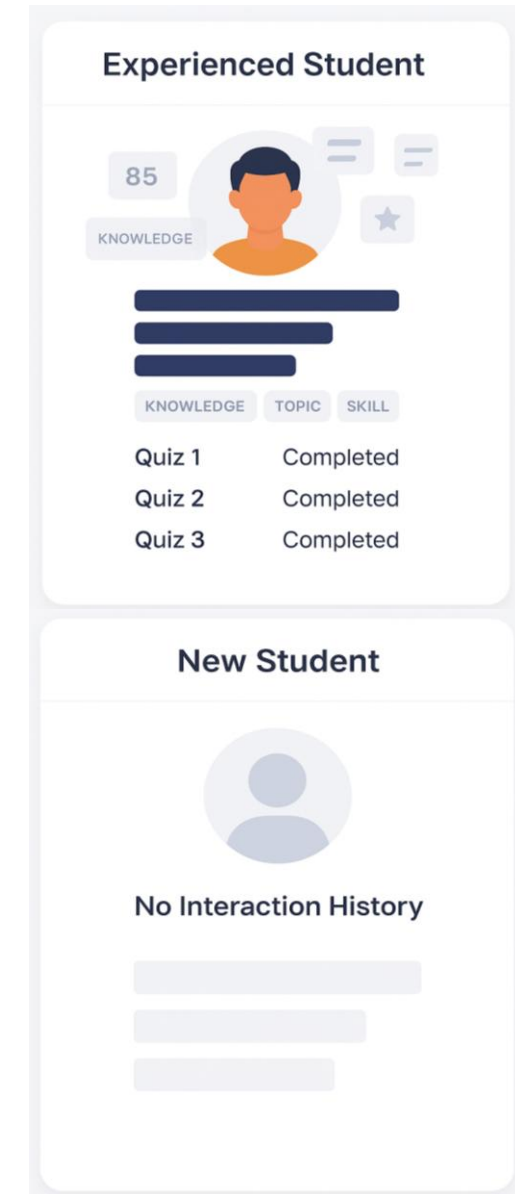
Knowledge Tracing



An example scenario for knowledge tracing in an Intelligent Tutoring System

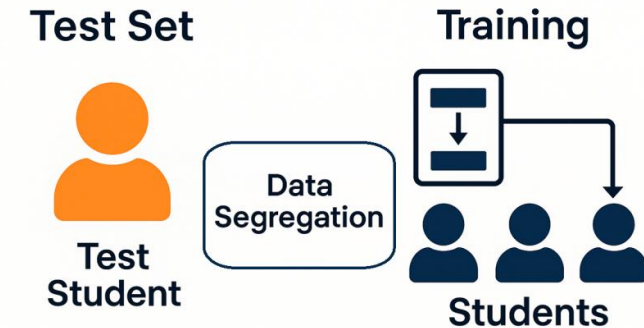
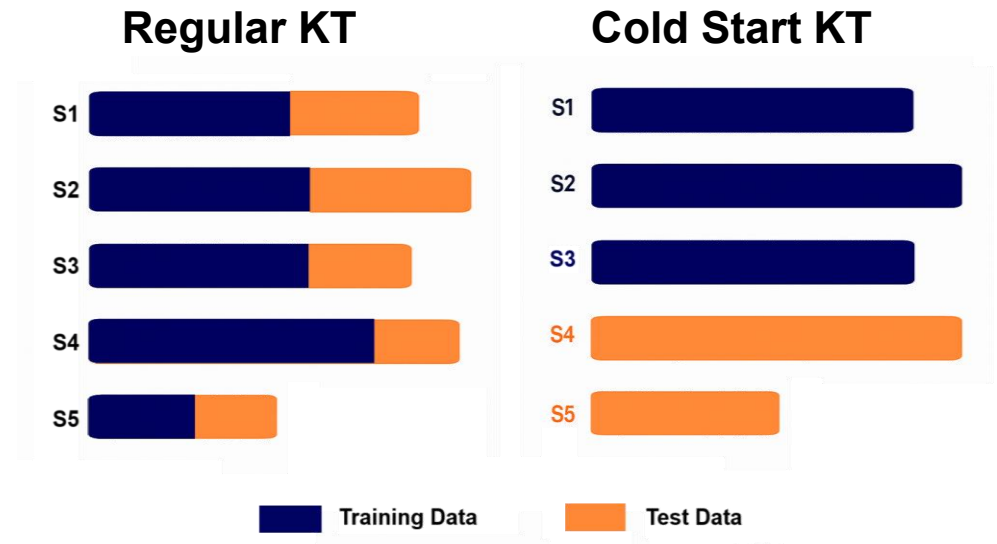
Understanding Cold Start in Knowledge Tracing

- A major challenge: Cold Start Problem
- Refers to
 - The challenge of accurately predicting a new student's knowledge state
 - When the system has minimal or no prior interaction data about them
- This delays effective personalization in Intelligent Tutoring Systems.
- Real-world systems often face this with brand-new learners logging in for the first time.



Prior Cold Start Work

- Previous cold start work
 - ✓ Trains on historical students.
 - ✓ Tests only on completely unseen learners.
 - ✓ Evaluation over multiple interactions.

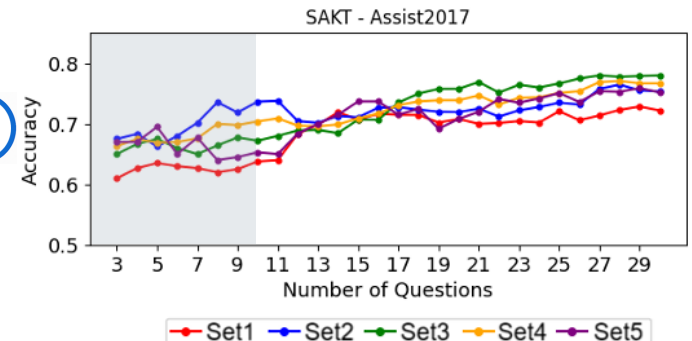
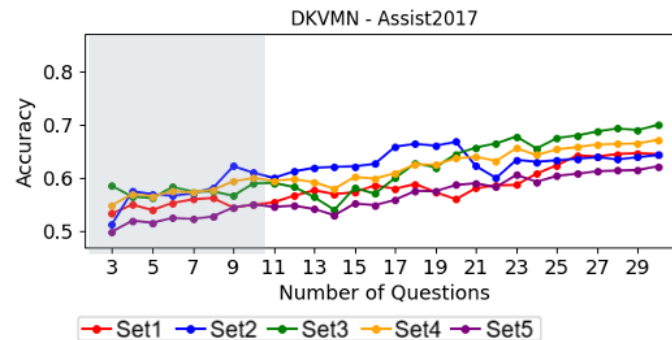
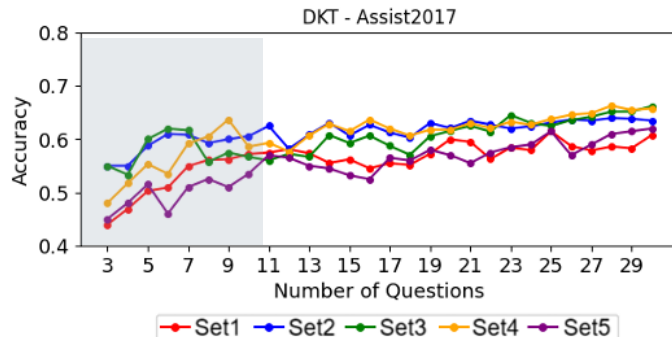
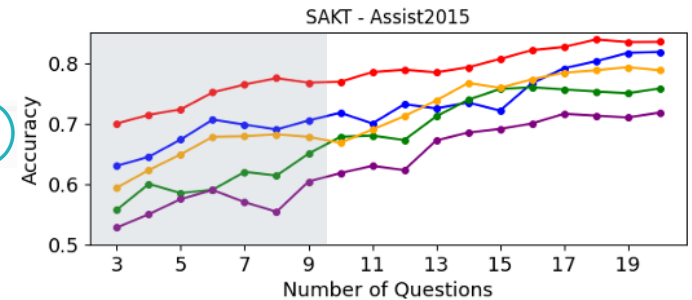
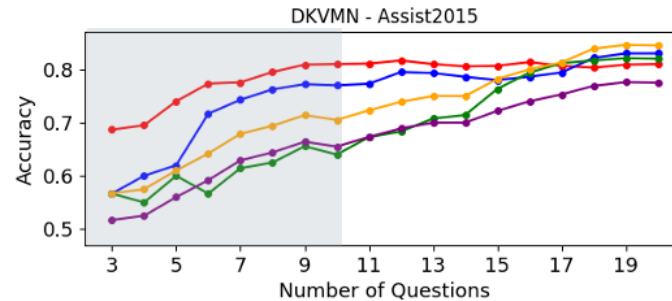
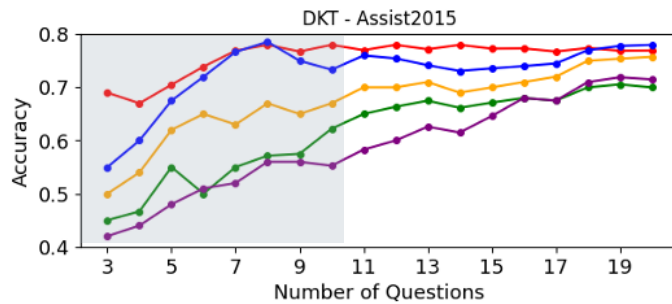
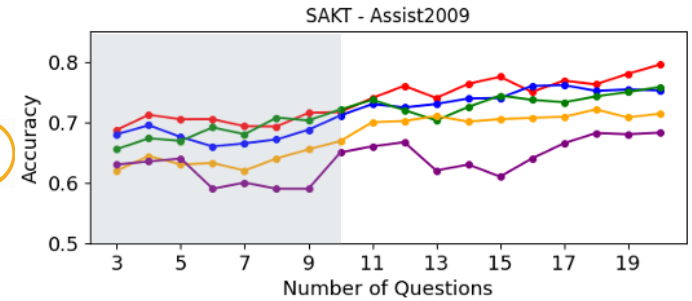
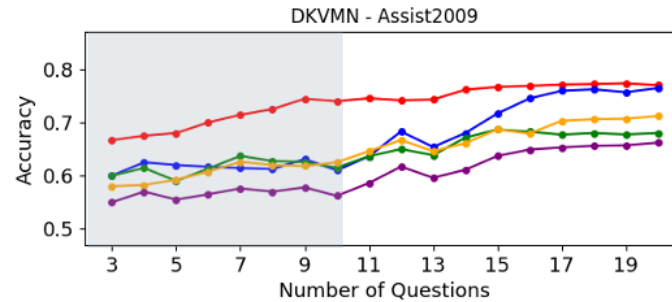
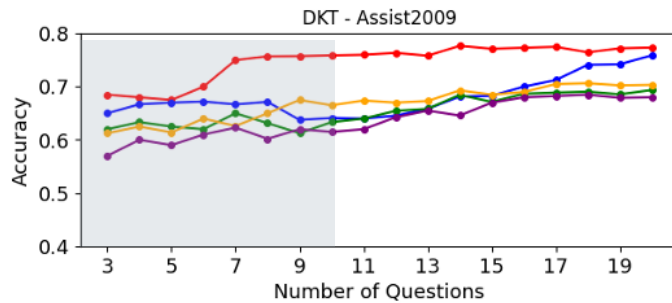


Results of Prior Work

ASSIST
2009

ASSIST
2015

ASSIST
2017



Intuition for Meta Learning

- ERM-trained KT models learn what works on average, but they do not adapt quickly to a new student with only a few early responses.
- This is similar to cold-start recommendation, where a system must serve a new user before enough interaction history is available.
- Meta-learning is useful because it learns an initialization that can adapt rapidly from a few examples instead of relying only on one fixed global model.
- Model-Agnostic Meta Learning (MAML), this algorithm identifies an ideal *"starting point"* for a model's settings, allowing it to reach peak performance after just one or two small updates.

MAML-KT

MAML-KT: Model-Agnostic Meta Learning for Knowledge Tracing

- **MAML-KT** applies meta-learning to knowledge tracing by treating each student's unique learning journey as an individual task.

Research Questions

1

Does meta learning improve early-phase new student performance over ERM baselines?

2

Does it scale with more new students?

(Cohort size from 10 students to 20 and 50 students)

3

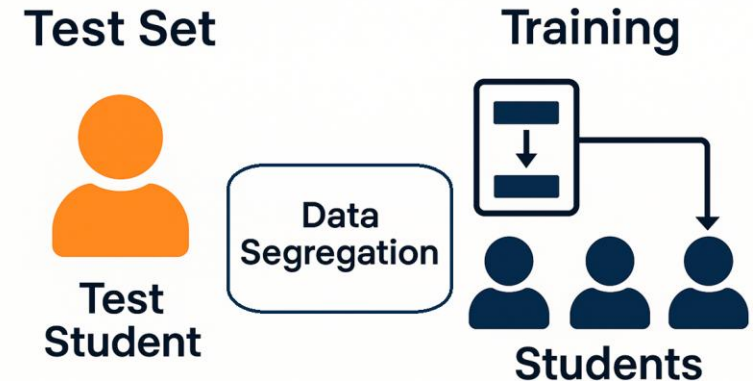
Under which sequence and content conditions does MAML-KT trail other ERM baselines?

(How do context length and number and timing of skills first appearing in the query related)

Experiment Setup

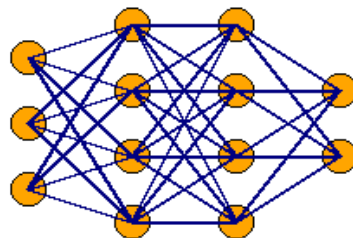
Ensuring True Cold Start

- Each dataset split into training set and 5 test sets (10, 20 and 50 students each).
- Test students' data completely removed from training phase.
- Ensures no prior knowledge leakage.
- Simulates real-world scenario: system faces a completely new learner.
- Evaluated performance as students answered 3 to 20 (or 30) questions.

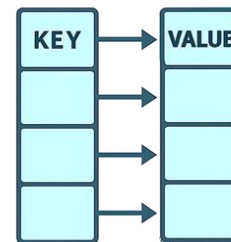


Baselines and Datasets Used

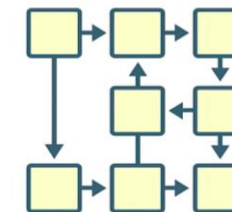
3 Different
State-of-the-art
Models



DKT
Deep Knowledge
Tracing
(RNN Based)



DKVMN
Dynamic Key-Value
Memory Network
(Memory Network Based)



SAKT
Self-Attentive
Knowledge Tracing
(Transformer Based)

3 Different
Concept-rich
Datasets



ASSISTMENT 2009
(Large number of Skills)

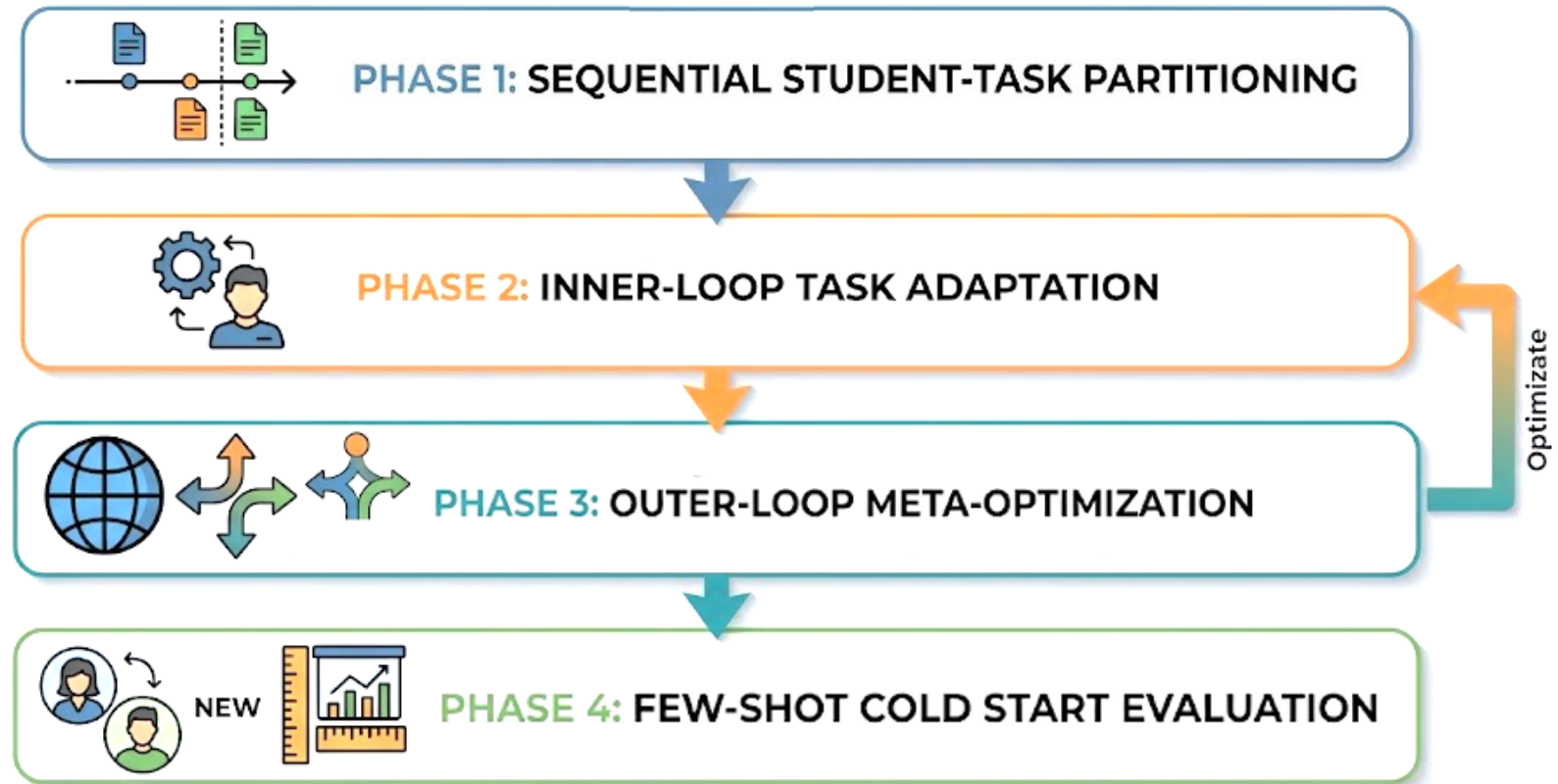


ASSISTMENT 2015
(Large Number of Students)



ASSISTMENT 2017
(Large Number of Interactions)

Methodology



Algorithm

Phase 1: Sequential Student-Task Partitioning

Phase 2: Inner-Loop Task Adaptation

Phase 3: Outer-Loop Meta-Optimization

Phase 4: Few-Shot Cold-Start Evaluation

Phase 1: Sequential Student-Task Partitioning

- Sample a student task
- Split the sequence into support and query
- Use the first K interactions as support
- Use the remaining interactions as query

Algorithm

Phase 1: Sequential Student-Task Partitioning

Phase 2: Inner-Loop Task Adaptation

Phase 3: Outer-Loop Meta-Optimization

Phase 4: Few-Shot Cold-Start Evaluation

Phase 2: Inner-Loop Task Adaptation

- Copy shared model parameters into student-specific fast parameters
- Compute support loss
- Perform inner gradient updates on the support set

Algorithm

Phase 1: Sequential Student-Task Partitioning

Phase 2: Inner-Loop Task Adaptation

Phase 3: Outer-Loop Meta-Optimization

Phase 4: Few-Shot Cold-Start Evaluation

Phase 3: Outer-Loop Meta-Optimization

- Compute query loss using the adapted parameters
- Accumulate query losses across the meta-batch
- Update the shared model parameters
- Use second-order gradients through the adaptation step

Algorithm

Phase 1: Sequential Student-Task Partitioning

Phase 2: Inner-Loop Task Adaptation

Phase 3: Outer-Loop Meta-Optimization

Phase 4: Few-Shot Cold-Start Evaluation

Phase 4: Few-Shot Cold-Start Evaluation

- Take a held-out new student
- Adapt using only the first few interactions
- Predict the remaining responses

Results

> Datasets

Assist 2009

Assist 2015

Assist 2017

> Cohort Size

10

20

50

> Cold Start Zone

Critical (First 10 Questions)

Moderate (11-15 Questions)

> Sets

5 for each zone

> Model Comparison

DKT (D)

DKVMN (K)

SAKT (S)

MAML-KT (M)

> Evaluation Metric

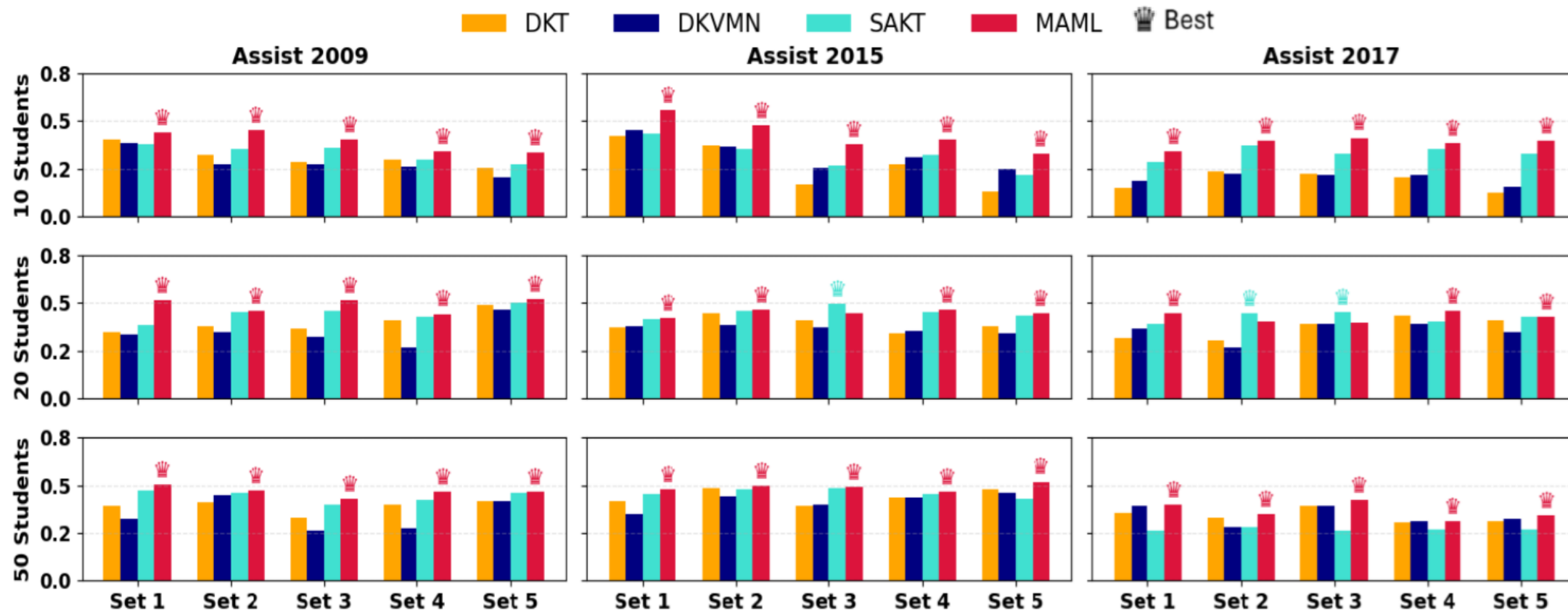
Average accuracy of prediction
in that zone

Critical Cold Start						Moderate Cold Start				
Dataset	Set 1	Set 2	Set 3	Set 4	Set 5	Set 1	Set 2	Set 3	Set 4	Set 5
10 New Students										
ASSIST 2009	75.3 ^M	75.9 ^M	72.3 ^M	67.2 ^M	66.8 ^M	79.4 ^M	78.1 ^M	76.6 ^M	71.9 ^M	70.5 ^M
	72.0 _D	68.1 _S	68.8 _S	63.9 _S	61.6 _S	76.6 _D	73.3 _S	72.6 _S	70.4 _S	64.7 _D
ASSIST 2015	84.5 ^M	78.1 ^M	70.0 ^M	72.1 ^M	66.0 ^M	88.3 ^M	78.7 ^M	76.8 ^M	77.9 ^M	75.6 ^M
	76.1 _N	69.8 _D	61.2 _S	65.7 _S	59.9 _D	81.0 _N	78.5 _N	71.2 _S	74.9 _N	69.7 _N
ASSIST 2017	67.4 ^M	71.9 ^M	72.9 ^M	70.6 ^M	71.7 ^M	69.1 ^M	72.8 ^M	69.4 ^M	72.7 ^M	73.0 ^M
	62.7 _S	69.9 _S	66.5 _S	68.2 _S	66.3 _S	68.9 _S	71.4 _S	69.0 _S	70.2 _S	69.7 _S
20 New Students										
ASSIST 2009	81.1 ^M	76.8 ^M	81.0 ^M	75.5 ^M	81.9 ^M	82.6 ^M	80.9 ^M	82.7 ^M	82.0 ^M	84.6 ^M
	70.8 _S	76.2 _S	76.9 _S	74.0 _S	80.3 _S	77.9 _S	80.5 _S	78.0 _S	79.1 _S	81.8 _S
ASSIST 2015	73.6 ^M	77.3 ^M	79.5 ^S	77.3 ^M	76.0 ^M	76.8 ^M	80.7 ^M	78.0 ^M	81.7 ^M	80.1 ^M
	73.4 _S	76.8 _S	75.9 _M	75.9 _S	74.7 _S	75.3 _D	79.1 _D	76.1 _S	74.3 _D	73.2 _S
ASSIST 2017	75.7 ^M	75.8 ^S	76.3 ^S	65.1 ^M	67.6 ^M	72.5 ^M	69.9 ^S	71.7 ^S	78.4 ^M	73.8 ^M
	71.3 _S	72.1 _M	71.9 _M	74.7 _D	74.2 _S	68.3 _S	68.8 _M	70.3 _M	73.9 _D	72.0 _D
50 New Students										
ASSIST 2009	80.1 ^M	77.8 ^M	74.4 ^M	77.1 ^M	77.5 ^M	85.0 ^M	78.4 ^M	72.0 ^M	79.6 ^M	77.2 ^M
	77.9 _S	76.7 _S	71.7 _S	73.9 _S	76.7 _S	81.3 _S	77.9 _S	70.9 _S	79.3 _S	75.5 _S
ASSIST 2015	78.2 ^M	79.9 ^M	79.3 ^M	77.2 ^M	81.5 ^M	79.7 ^M	81.8 ^S	82.6 ^M	80.1 ^M	80.2 ^M
	76.5 _S	78.9 _D	78.6 _S	76.3 _S	78.3 _D	79.4 _S	81.4 _M	81.0 _S	77.8 _S	76.8 _S
ASSIST 2017	71.9 ^M	67.6 ^M	74.0 ^M	65.1 ^M	67.6 ^M	71.4 ^M	69.7 ^M	73.0 ^M	65.6 ^M	68.1 ^M
	71.2 _N	66.6 _D	71.5 _N	64.9 _N	66.1 _N	69.6 _N	67.1 _D	69.6 _D	64.9 _N	65.0 _D

* M: MAML, D: DKT, N:DKVMN, S: SAKT

Results (Critical Cold Start)

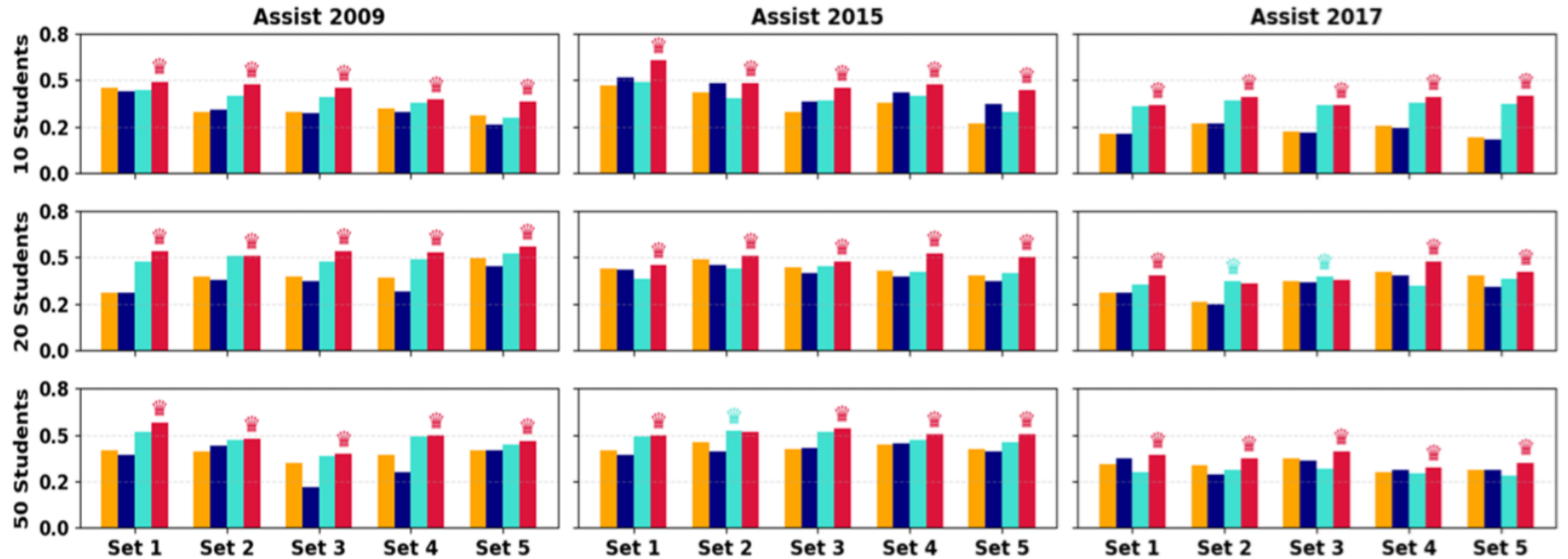
Critical Cold-Start (Questions 3-10) Average Accuracy Across Datasets - 5 Sets x 4 Models x 3 New Student Set Sizes



Results (Moderate Cold Start)

Moderate Cold-Start (Questions 11-15) Average Accuracy Across Datasets - 5 Sets x 4 Models x 3 New Student Set Sizes

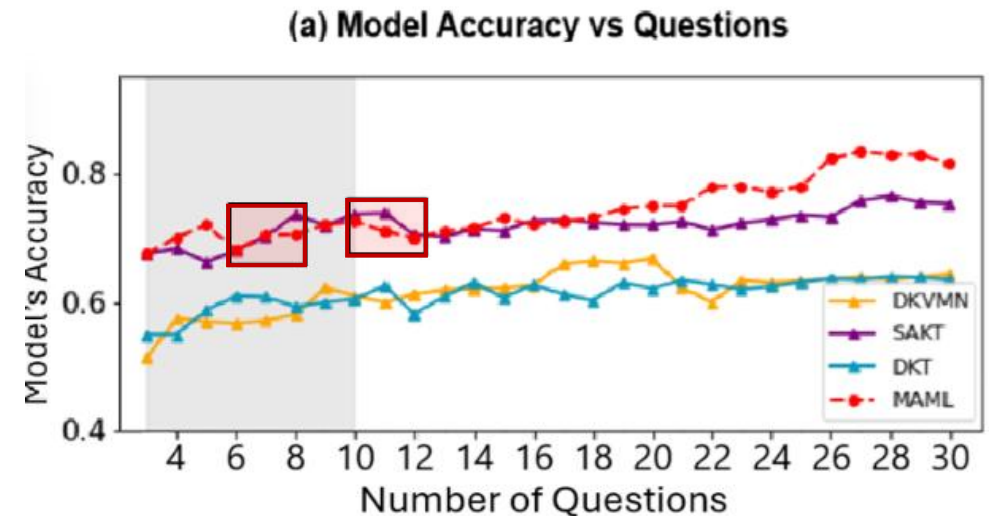
DKT DKVMN SAKT MAML 🏆 Best



Results (Edge Cases)

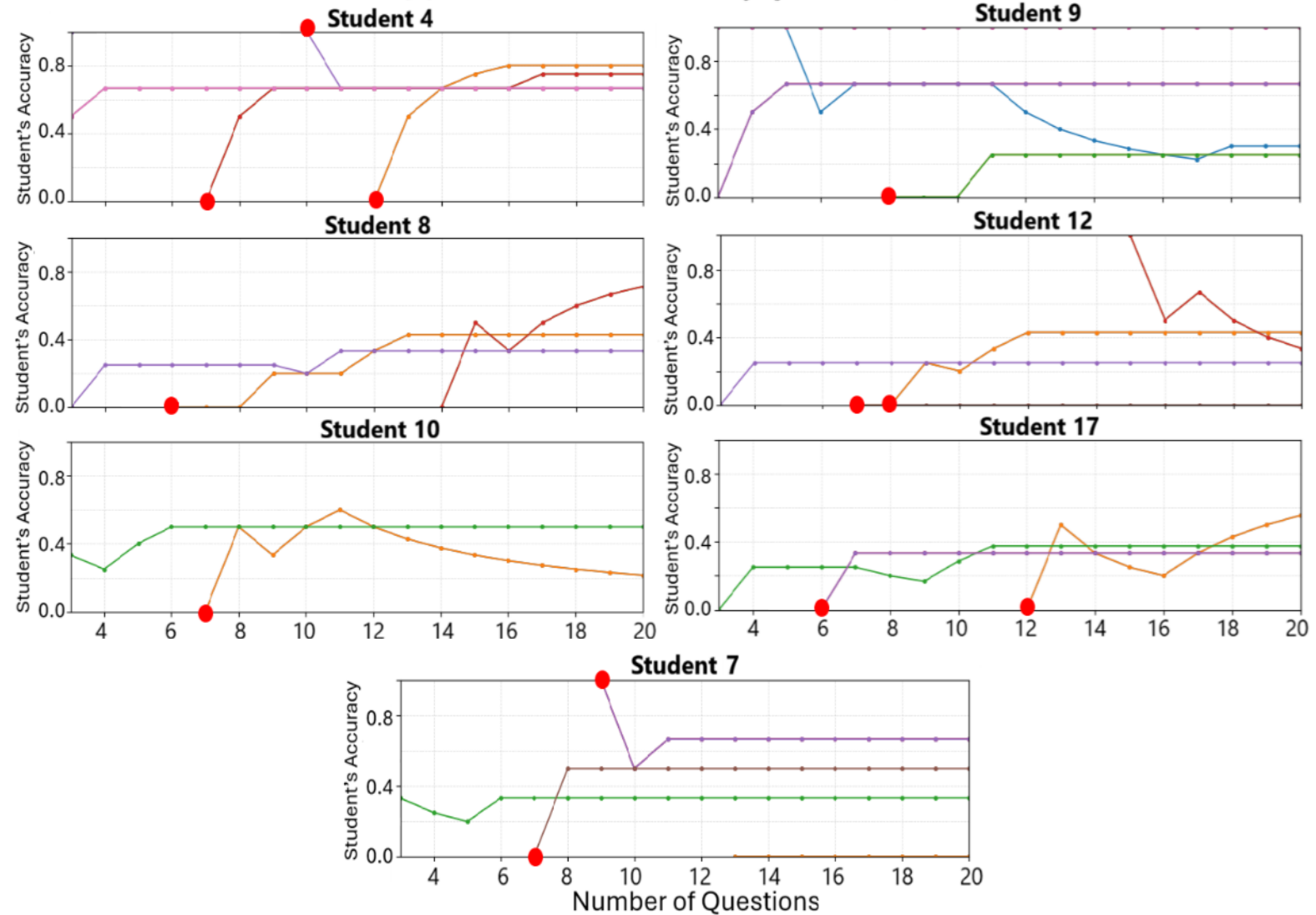
- MAML-KT briefly falls behind SAKT for some of the points.
- Cause: Frequent new skills appear in query but not support.
- This creates a short support–query mismatch.
- The dip is localized, not a general failure.

Assist2017 - 20 New Students
Critical and Moderate scenarios



Results (Edge Cases)

(b) Per Student Answer Accuracy by Skill vs. Questions



What Did We Learn?

Reflections on the Research Questions

- **RQ1: Does Meta-Learning improve cold start?**
 - MAML-KT achieves the best early accuracy for new students.
 - Confirms that learning to adapt is stronger than using a fixed global model.
- **RQ2: Does it scale with more new students?**
 - Gains remain consistent from 10 $\rightarrow$ 20 $\rightarrow$ 50 student cohorts.
 - Shows that MAML-KT is not only effective in small test sets but also robust at larger scale.
- **RQ3: When does MAML-KT trail baselines?**
 - SAKT can briefly catch up when new skills appear suddenly and more frequently in the query.
 - Suggests that meta-learning is highly effective, but skill novelty and heterogeneity still create short-term challenges.

“Cold start is not solved by stronger global models alone, it is improved by models that can adapt quickly to each new learner.”

Future Work

- Extend to other cases of cold-start settings
- Explore skill-level task construction
- Handle frequent new-skill arrivals

Conclusion

- MAML-KT successfully addresses the cold-start problem by framing student knowledge tracing as a few-shot adaptation task from very limited interactions.
- The model significantly outperforms standard baselines in early-phase prediction accuracy
- Maintaining its effectiveness and showing that increasing the student cohort size also results in strong performance.
- Future research will focus on making adaptation strategies more robust for skill transitions and knowledge component changes.

Thank you!

For any more questions, please send me an email

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